

SIYI CHEN

celine072022@gmail.com | <https://www.linkedin.com/in/celine-c-04854725b/> | <https://github.com/celinec41>

EXPERIENCE

Nomura

Shanghai, China

Asset Management Data & Operations Intern

May 2025 – Aug 2025

- Built automated Python data pipelines and API integrations to extract, clean, and validate 2,500+ daily fund records across 3 funds, replacing manual work and cutting preparation time by 75%.
- Owned data quality across multiple sources, tracing missing records and abnormal values to their root cause and fixing them before reports went out.
- Calculated 15 performance and risk measures for each fund and delivered them in regular reports used by the team.
- Built a Python tool that automatically pulls currency and gold prices and creates comparison charts, cutting data lookup time by 80%.

PROJECTS

Natural Language SQL Query Generator

Tools: Python, SQL, SQLite, Tkinter, Groq API (Llama 3.3)

- Built a desktop app that turns plain-English questions into working SQL queries using the Llama 3.3 AI model, answering questions across a multi-table database with 15,000+ records.
- Added a validation layer that checks every generated query against the database schema and runs it read-only, blocking all 35 test attempts to change or delete data.

Financial Impact Prediction Model — SDSS Datathon

Tools: Python, pandas, scikit-learn, Streamlit

- Cleaned and modeled 16,000+ national survey responses to predict how household finances respond to an economic shock, performing about 1.5x better than random guessing.
- Found that high credit card and other unsecured debt were the top risk factors, and shared the insights through an interactive Streamlit app.

Global Ecological Footprint Dashboard

Tools: Python, pandas, Dash, Plotly

- Cleaned and joined multi-country ecological footprint and biocapacity datasets, then built an interactive dashboard that lets users explore and compare country-level trends.

Dishly — AI Restaurant Menu Agent (Battle of the Schools: UofT vs. Waterloo, Honourable Mention)

Tools: Next.js, Claude API, Steel, Git

- Built an AI web agent with a team of 4 that reads a restaurant's menu, explains each dish with a photo, and recommends dishes backed by customer reviews.
- Built the menu parser and a cache that reuses lookups across restaurants, plus a review analyzer that scores customer sentiment for each dish.

EDUCATION

University of Toronto (St. George Campus)

Toronto, ON

Honours BSc, Computer Science Specialist (Statistics Minor), ASIP Co-op

Sep 2024 – Jun 2029

- Recipient of the University of Toronto International Scholar Award, valued at up to \$100,000
- Relevant Coursework: Databases, Data Structures and Algorithms, Probability and Statistics, Machine Learning, Software Design

SKILLS

Languages: SQL, Python (pandas, NumPy), Java, C

Data Engineering: Data Pipelines (ETL), Relational Data Modeling, Data Validation & Quality, API Integration, SQLite

Analytics & Tools: Dashboards (Dash, Plotly, Streamlit), scikit-learn, Excel, Jupyter, Git/GitHub